

Mayank Singh

mayank.goa@gmail.com — 814-418-8274 — linkedin.com/in/mayank-singh-011536230

EDUCATION

Pennsylvania State University, State College, PA
Bachelor of Science in Computer Science

December 2025

Relevant Courses: Data Structure and Algorithms, Database Management Systems, Operating Systems Design, Machine Learning.

PROFESSIONAL SUMMARY

Detail-oriented Computer Science student at Penn State with hands-on experience in software development, AI research, and backend systems. Skilled in Python, Java, SQL, and modern development frameworks. Passionate about applying AI for real-world impact with a strong focus on ethical design and user-centric systems.

SKILLS

- **Programming:** Python, Java, C, C++, SQL
- **Frameworks and Tools:** Streamlit, Docker, Spring Cloud, REST APIs, GitHub, Apache
- **Databases:** MySQL, SQL Server.
- **Software Development:** Test-driven development, application deployment, debugging.
- **Soft Skills:** Analytical thinking, effective communication, team collaboration.

EXPERIENCE

Penn State Research — ReSENSE Lab

State College, PA

May 2025 – Present

- Developing a GenAI-powered platform for generating personalized content across education, marketing, and support domains.
- Leveraging multimodal AI (language + vision models) to automate creative workflows and optimize real-time user interaction.
- Ensuring ethical AI practices through transparency, human-in-the-loop feedback, and sector-specific customization.
- **Skills:** Python, Streamlit, Generative AI, System Architecture, UI/UX

Research Assistant – Hybrid AI for Early Diabetes Detection

State College, PA

Jan 2025 – May 2025

- Designed a dual-model system using Random Forest and ACT-R for diabetes risk prediction.
- Built an interactive Streamlit app comparing ML and cognitive predictions with visual outputs.
- Integrated fairness-aware logic, case-based reasoning, and ethical diagnostic principles.
- **Skills:** Streamlit, Python, ACT-R, Random Forest (ML)

PROJECTS

Automated Job Application System

- **Technologies Used:** n8n, GPT-4.0 Mini, Ampify Web Scraper, Google Sheets API
- Built an automation pipeline to scrape job listings from Indeed using Ampify actors and filter them via GPT-based relevance.
- Generated personalized cover letters using GPT and automatically applied it to Google Sheets.
- Tracked applications and status updates in Google Sheets using dynamic n8n workflows.

AI WhatsApp Chatbot

- **Technologies Used:** n8n, GPT-4.0 Mini, WhatsApp Cloud API, Webhooks
- Created an AI-powered chatbot capable of dynamic and contextual replies to WhatsApp messages along with describing images.
- Enabled session-aware interactions and real-time communication using GPT and Messenger Send node.

Audio-to-Image Generation App

- **Technologies Used:** Python, Streamlit, OpenAI Whisper, DALL·E
- Developed a Streamlit app that converts spoken audio into AI-generated images using Whisper for transcription and DALL·E for image synthesis.

Image Caption Generator

- **Technologies Used:** Python, Hugging Face (BLIP), Streamlit
- Built an AI tool that generates descriptive captions for images using BLIP and Hugging Face Transformers.
- Deployed a responsive Streamlit interface for real-time caption output.

University Course Registration System

- **Technologies Used:** SQL, PostgreSQL, Python
- Developed a course registration system with functionalities for course selection, enrollment, and conflict checking.
- Included features for transactional integrity, report generation, and rollback handling.

Inventory Management Microservices System

- **Technologies Used:** REST API, Docker
- Designed a microservices-based inventory system where services handled cataloging, orders, and notifications.
- Ensured inter-service communication via REST APIs and deployed using Docker containers.